BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Network Hardware and Topologies

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What distinguishes a WAN?

2. Why use shared computing standards?

A) Every device needs one maker
B) Different products work together
C) No communication rules are needed
D) All hardware becomes identical

3. "SMTP's role is synchronising folders and read status across mail clients." Correct this. Explain why.

4. A star LAN has one central switch. "One broken workstation cable stops the whole network." Correct this. Compare a broken cable with a failed switch.

Today's objectives

Explain the core ideas of network hardware and topologies using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

NIC
Provides a device's connection to a network.

Switch
Forwards frames between devices within a LAN using MAC address information.

Router
Connects networks and forwards packets using IP address/routing information.

Star and mesh
Star connects devices via a central point; mesh offers multiple potential paths between nodes.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

NIC	Provides a device's connection to a network.
Switch	Forwards frames between devices within a LAN using MAC address information.
Router	Connects networks and forwards packets using IP address/routing information.
Star and mesh	Star connects devices via a central point; mesh offers multiple potential paths between nodes.

I Do - One worked example

Explain how data travels from a classroom PC to a server on another network.

1. The PC connects via its NIC.
2. A switch forwards traffic inside the LAN.
3. A router forwards the packet to the other network.
4. The destination network delivers data to the server.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Compare star and mesh topology after one connection fails.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the number of links in a fully connected mesh of five computers.

```
devices = 5
links = devices * (devices - 1) DIV 2
print(links)
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. State what a switch does in a LAN. [2 marks]

Q3. Give the purpose of a router. [2 marks]

Q4. Calculate full-mesh connections for four nodes. [2 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A switch normally forwards using IP routes between networks. ☐ T ☐ F
2. Provides a device's connection to a network. ☐ T ☐ F
3. Forwards frames between devices within a LAN using MAC address information. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 A WAN links networks over a large area, often using links provided by other organisations. ☐

Q2 B - Different products work together ☐

Q3 Sending email from a client to a mail server and between mail servers. ☐

Q4 A broken workstation cable usually affects just that device. If the central switch fails, devices using it lose contact. Available paths determine which connections still work. ☐

Code First - Model

```
devices = 5
links = devices * (devices - 1) DIV 2
print(links)
```

Output: 10

We Do - Check the model

1. Explain the trade-off: mesh uses more connections and costs more. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Forwards frames to the appropriate device/port using learned MAC address information. ☐

Q3 Connects separate networks and forwards packets between them using IP routing. ☐

Q4 4 x 3 / 2=6 connections. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: